# MUNTASIR MAHDI

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## SUMMARY


- PhD student in Electrical Engineering with 3+ years of hands-on cleanroom microfabrication experience (photolithography, PECVD, ALD, ICP-RIE, sputtering)
- Experienced in process development, thin-film deposition, and etch optimization for semiconductor device prototyping
- Strong background in experimental design, data analysis (Python/MATLAB), and device characterization (SEM, AFM, ellipsometry)


## EDUCATION

**Doctor of Philosophy,** (GPA: 3.9 out of 4) Aug 2024 — Jul 2028
*Electrical and Computer Engineering – Virginia Tech, VA, USA*

**Master of Science,** (GPA: 3.5 out of 4) May 2021 — Aug 2024
*Electrical and Computer Engineering – Auburn University, AL, USA*

**Bachelor of Science,** (GPA: 3.58 out of 4) Mar 2015 — Apr 2019
*Electrical and Electronic Engineering – Shahjalal University of Science and Technology, Bangladesh*

## TECHNICAL SKILLS

**Microfabrication:** Photolithography, E-beam lithography, Lift-off, Wet chemistry processing, Thin-film deposition: PECVD, ALD, DC/RF sputtering, Dry etching (ICP-RIE, Ion milling), Process development & optimization, cleanroom processing.
**Characterization:** SEM, AFM, XRD, Spectroscopic Ellipsometry, Oscilloscope, VNA, QD-PPMS, Keithley instruments
**Simulation & Design:** Keysight ADS, Cadence Virtuoso, Altium, Fusion 360, Synopsys Sentaurus TCAD.
**Programming & Automation:** Python (data analysis, visualization, automation), C, MATLAB, OriginLab, Microcontroller, Raspberry Pi.

## RESEARCH & PROFESSIONAL EXPERIENCE

**Publications:** 9 journal papers, 3 conference proceedings (IEEE, APS, Phys. Rev. Applied)

**Graduate Research Assistant**, Electrical Engineering, Virginia Tech Aug 2024 — Present
- Characterize MBE-grown Ge/GeSn thin films using ellipsometry, AFM, and XRD to evaluate thickness, strain, and defect density
- Correlate material properties with CMOS-compatible fabrication requirements and device performance
- Analyze process–structure–property relationships to guide material and device optimization

**Graduate Research Assistant**, Electrical Engineering, Auburn University May 2021 — Aug 2024
- Fabricated micro/nano devices using photolithography, e-beam lithography, PECVD, sputtering, and ICP-RIE dry etching.
- Developed and optimized thin-film deposition processes (Nb, dielectrics)
- Performed metal deposition, lift-off, and pattern transfer; resolved fabrication defects (residue, roughness, adhesion issues)
- Optimized dry etching (ICP-RIE) recipes for anisotropy, selectivity, and surface quality
- Designed and built a high-vacuum ion milling system for thin-film processing and surface modification
- Conducted failure analysis and process troubleshooting to improve device yield and reproducibility
- Developed a Python-based automation tool, accelerating experimental iteration, saving $4000+ in software cost

**Undergraduate Research Assistant**, Electrical Engineering, Shahjalal University of Science and Technology
- Transistor Modeling: Modeled and analyzed CNTFET I–V characteristics using MATLAB/Simulink. Jul 2018 — Jul 2019
- Mars Rover Robot: Built motor control circuits, mechanical chassis for a Mars rover 2015 — 2017

## INDUSTRIAL TRAINING

**Industrial Training on IC Layout Design**, Ulkasemi Inc., Bangladesh Feb 2020
- IC layout design using Cadence Virtuoso (standard cells, hierarchical layout)

**Industrial Technology in Elec. Engineering & Instrumentation**, Training Institute of Chemical Industries, Bangladesh Dec 2018
- Trained in electrical safety, power generation & distribution, generators, transformers, motors, sensing, and control.

## TEACHING & MENTORSHIP

**Graduate Teaching Assistant,** Department of ECE, Virginia Tech Aug 2024 — Jul 2028
- Mentored student projects involving semiconductor device fabrication and simulation (MOSCAP, Ge-based devices)
- Supported VLSI design and debugging using Cadence Virtuoso.

**Graduate Teaching Assistant**, Department of ECE, Auburn University Jan 2024 — May 2024
- Supported a 75-student course in Electrical Engineering fundamentals; led review sessions and provided exam support.

**Lecturer,** Department of EEE, University of Science & Technology Chittagong (USTC), Bangladesh Dec 2019 — May 2021
- Taught courses in Electrical Circuits, Electronics, and Control Systems with corresponding labs.